

↗ a random variable

$$\hat{\mathcal{M}}_n = \frac{S_n}{n} \quad , \quad S_n = \sum_{i=1}^n X_i \quad , \quad \{X_i\} : \text{i.i.d, mean } \mathcal{M}, \text{ variance } \sigma^2$$

$$E[\hat{\mathcal{M}}_n] = \mathcal{M} \quad , \quad \text{var}(\hat{\mathcal{M}}_n) = \frac{\sigma^2}{n}$$

* Consider $Y_n = \frac{\hat{\mathcal{M}}_n - \mathcal{M}}{\sqrt{\frac{\sigma^2}{n}}}$ $= \frac{\hat{\mathcal{M}}_n - \mathcal{M}}{\frac{\sigma}{\sqrt{n}}}$

centering and scaling
standard dev.

$$E[Y_n] = 0 \quad , \quad \text{var}(Y_n) = \frac{1}{(\text{sd})^2} \cdot \text{var}(\hat{\mathcal{M}}_n - \mathcal{M}) = \frac{n}{\sigma^2} \cdot \text{var}(\hat{\mathcal{M}}_n) = 1$$

Y_n has mean 0 and variance 1 irrespective of X .

∴ What is $F_{Y_n}(\cdot)$? ↗ distribution of Y_n

↙ CDF

$$\lim_{n \rightarrow \infty} F_{Y_n}(\cdot) = \Phi(\cdot) = F_{N(0,1)}(\cdot)$$

∴ Y_n converges to $N(0,1)$ in distribution.

standard normal

• CENTRAL LIMIT THEOREM

Let $\{X_n, n \geq 0\}$ denote a sequence of i.i.d random variables each with mean $\mathcal{M}$ and variance $0 < \sigma^2 < \infty$.

Denote $\hat{\mathcal{M}}_n = \frac{\sum_{i=1}^n X_i}{n}$ and $Y_n = \frac{\hat{\mathcal{M}}_n - \mathcal{M}}{\frac{\sigma}{\sqrt{n}}}$.

Then, Y_n converges to $N(0,1)$ in distribution.

Q: What if $E\{X_i\} = 0$ and $\text{var}(X_i) = 1$?

ans: $Y_n = \frac{S_n}{\sqrt{n}}$

$$\frac{S_n}{n} \xrightarrow{\text{a.s.}} \mathcal{M} \quad \Bigg| \quad \frac{S_n}{\sqrt{n}} \xrightarrow{d} N(0,1)$$

∴ For large enough n , $Y_n \sim N(0,1)$
 $\mathcal{M}_n \sim N\left(\mathcal{M}, \frac{\sigma^2}{n}\right)$

$P(S_n < c) = \Phi(z) \quad \text{where} \quad z = \frac{c - n\mathcal{M}}{\sigma\sqrt{n}}$

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PROOF: $Y_n = \frac{\hat{\mathcal{M}}_n - \mathcal{M}}{\frac{\sigma}{\sqrt{n}}} \times n = \frac{S_n - n\mathcal{M}}{\sigma\sqrt{n}}$

$P(S_n < c) = P\left(Y_n < \frac{c - n\mathcal{M}}{\sigma\sqrt{n}}\right) = F_{Y_n}\left(\frac{c - n\mathcal{M}}{\sigma\sqrt{n}}\right) = \Phi\left(\frac{c - n\mathcal{M}}{\sqrt{n}\sigma}\right)$

• Random Vectors

$$\bar{X} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix} \quad / \quad \bar{X} = (X_1, \dots, X_n)^T$$

Components X_i are scalar valued random variable defined on the same space $(\Omega, \mathcal{F}, P)$.

∴ they may be correlated.

e.g. $\bar{X} = (X_1, X_2)^T$, $X_1 = Z_1$, $X_2 = Z_1 + Z_2$
 Z_1 and Z_2 are independent standard normal.

CDF: $F_{\bar{X}}(\bar{x}) = F_{X_1, \dots, X_n}(x_1, \dots, x_n)$	joint CDF
PDF: $f_{\bar{X}}(\bar{x}) = f_{X_1, \dots, X_n}(x_1, \dots, x_n)$	joint PDF

$$E[\bar{X}] = (E[X_1], \dots, E[X_n])^T$$

$\bar{Y} = A\bar{X} + b$ ↗ vector $\Rightarrow E[\bar{Y}] = A \cdot E[\bar{X}] + b$

↙ matrix

• COVARIANCE MATRIX

$$\begin{matrix} X_1 & X_2 & \dots & X_n \\ X_1 & \begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \dots \end{pmatrix} \\ X_2 & \\ \vdots & \\ X_n & \begin{pmatrix} \dots \dots \dots \text{Var}(X_n) \end{pmatrix} \end{matrix}$$

a symmetric matrix

variances along diagonal

other elements are covariances

$$C_{\bar{X}} = E[(\bar{X} - E[\bar{X}])(\bar{X} - E[\bar{X}])^T]$$

* C_X is always positive semi-definite.
 i.e. For any vector $a \neq 0$, $a^T C_X a \geq 0$
 PROOF: Let $u = a^T(\bar{X} - E[\bar{X}])$
 $a^T C_X a = E\{uu^T\} = E\{u^2\} \geq 0$

Q: PROVE:

If $\bar{Y} = A\bar{X} + b \Rightarrow C_{\bar{Y}} = AC_{\bar{X}}A^T$ and $f_Y(y) = \frac{1}{|\det(A)|} \cdot f_X(A^{-1}(y-b)) \Rightarrow Y = AX + b \Rightarrow X = A^{-1}(Y-b)$

Consider $Y = g(X)$ where g is monotone, continuous, differentiable.

$Y = AX \Rightarrow X = A^{-1}Y$
 $H(\cdot) = A^{-1}Y$

$$\Rightarrow f_Y(y) = f_X(h(y)) \cdot \left| \frac{dh}{dy}(y) \right| \quad \text{where } h = g^{-1}$$

Q: How do you apply this to vectors, i.e. $\bar{Y} = g(\bar{X})$?

$$\bar{Y} = G(\bar{X})$$

$$\bar{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} = \begin{pmatrix} G_1(X_1, \dots, X_n) \\ G_2(X_1, \dots, X_n) \\ \vdots \\ G_n(X_1, \dots, X_n) \end{pmatrix}$$

e.g. $\bar{Y} = \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = \begin{pmatrix} 2X_1 \\ X_1 + X_2 \end{pmatrix}$, $G_1(X_1, X_2) = 2X_1$, $G_2(X_1, X_2) = X_1 + X_2$

Let $H = G^{-1}$

$$X = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix} = \begin{pmatrix} H_1(Y_1, \dots, Y_n) \\ H_2(Y_1, \dots, Y_n) \\ \vdots \\ H_n(Y_1, \dots, Y_n) \end{pmatrix}$$

$$X_1 = H_1(Y_1, Y_2) = \frac{Y_1}{2}, \quad X_2 = H_2(Y_1, Y_2) = Y_2 - \frac{Y_1}{2}$$

$$f_Y(\bar{y}) = f_X(H(\bar{y})) |J|$$

$$J = \text{jacobian} = \begin{pmatrix} \frac{\partial H_1}{\partial y_1} & \frac{\partial H_1}{\partial y_2} & \dots & \frac{\partial H_1}{\partial y_n} \\ \frac{\partial H_2}{\partial y_1} & \frac{\partial H_2}{\partial y_2} & \dots & \frac{\partial H_2}{\partial y_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial H_n}{\partial y_1} & \frac{\partial H_n}{\partial y_2} & \dots & \frac{\partial H_n}{\partial y_n} \end{pmatrix}$$

The Jacobian gives the ratio of incremental areas $dx, dx_2, \dots, dx_n$ and $dy, dy_2, \dots, dy_n$.

$$Y = AX + b \Rightarrow CY = ACX + AT$$

PROOF: $C_Y = E[(Y - E[Y])(Y - E[Y])^T]$

$$= E[(AX + b - AE[X] - b)(AX + b - AE[X] - b)^T]$$

$$= E[(AX - AE[X])(AX - AE[X])^T]$$

$$= E[A(X - E[X])(X - E[X])^T A^T]$$

$$= A \cdot E[(X - E[X])(X - E[X])^T] \cdot A^T$$

$$= AC_X A^T$$